



Outcomes of ECT for depression in patients with and without eating disorders

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ABSTRACT

Eating disorders (ED) have a high comorbidity with major depressive disorder (MDD). Antidepressants often lack effect in these patients. While electroconvulsive therapy (ECT) is the most effective treatment for severe MDD, there are no large studies in the ED patient group yet. We aimed to compare the outcome of ECT for MDD in patients with and without eating disorders. We conducted a register-based study for patients with MDD and comorbid ED treated with ECT in Sweden between 2012 and 2023. The outcomes were compared to a matched control group without comorbid ED. The primary outcome was a response to treatment according to the Clinical Global Impression Improvement Scale (CGI-I). Secondary outcomes were remission according to CGI-I and subjective memory complaint. There were 861 patients in the ED group and 1722 in the control group; of these 93.2% were female. The response rate was 58.4% in the ED group and 62.4% in the control group (OR 0.8, 95% CI 0.7–0.9, $p < 0.05$). The frequency of subjective memory complaints in the ED versus the control group was 25.9% and 24.8%, respectively (OR 1.1, 95% 0.8–1.3 CI, $p = 0.6$). High response rates after ECT for MDD were found in the ED group but were significantly lower than in the matched control group. Subjective memory complaints did not differ significantly. Further studies are required comparing the outcome of ECT versus alternative treatments among patients with ED complicated by severe MDD.

1. Introduction

Eating disorders (ED) are serious and complex illnesses, that have recently increased in incidence (Calvo-Rivera et al., 2022; Crone et al., 2023). They are most common in young women, with a prevalence estimated at 5.7% in women and 2.2% in men (Galmiche et al., 2019; Silén and Keski-Rahkonen, 2022). There are several types of ED, including anorexia nervosa (AN), bulimia nervosa, binge eating disorder, other specified eating disorder, pica, rumination disorder, avoidant/restrictive food intake disorder, and unspecified eating disorder (American Psychiatric Association, 2013). AN is considered to have the highest mortality among all psychiatric disorders, where suicide is one of the main causes of death (Arcelus et al., 2011; Crone et al., 2023; Goldstein and Gvion, 2019).

Comorbid psychiatric disorders are frequent among ED patients, where major depressive disorder (MDD) is one of the most common (Blinder et al., 2006; Calvo-Rivera et al., 2022). Concurrent MDD has been shown to increase morbidity and mortality in ED patients (Franko and Keel, 2006; Milos et al., 2004). Furthermore, upon initial treatment and improvement in ED symptoms, patients may experience

deterioration in depression and suicidality (Shilton et al., 2020). Therefore, it is crucial to find an effective treatment. However, antidepressants often lack efficacy in this patient group (Leblé et al., 2017; Mischoulon et al., 2011). In low-weight patients, it has been hypothesized to be caused by an altered serotonin metabolism due to malnutrition (Ferguson et al., 1999).

While electroconvulsive therapy (ECT) is the most effective treatment alternative for severe MDD (Pagnin et al., 2004; UK ECT Review Group, 2003), there are no large, controlled trials in the ED patient group. Several small studies have shown significant improvement in MDD and ED symptoms after ECT, without severe adverse effects (Andersson et al., 2023; Poutanen et al., 2009; Shilton et al., 2020; van der Lelie et al., 2021). Because of this, it is of interest to investigate the treatment outcomes in larger populations.

This study aimed to compare the outcome of ECT for MDD in patients with eating disorders to a matched control group. There are several known positive and negative predictors of response for ECT in MDD. Higher age and MDD with psychotic symptoms are associated with higher remission rates (Diermen et al., 2018; O'Connor et al., 2001; Petrides et al., 2001). Psychiatric comorbidity with anxiety disorder, substance use, or personality disorder is associated with lower remission

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Abbreviations

ED –	Eating disorders
AN –	Anorexia nervosa
MDD –	Major depressive disorder
ECT –	Electroconvulsive therapy
Q-ECT –	Swedish National Quality Register for ECT
ICD-10 –	International Classification for Diseases and Related Health Problems – Tenth revision
CGI-I –	Clinical Global Impression Improvement
ADHD –	Attention Deficit Hyperactivity Disorder
CPRS –	Comprehensive Psychopathological Rating Scale
SMD –	Standardized Mean Difference

rates after ECT, as well as long antidepressant treatment pre-ECT and medication with benzodiazepines, lamotrigine, or anticonvulsants (Brus et al., 2017; Ferrea et al., 2024; Haq et al., 2015; Gilling et al., 2024). It is uncertain if electrode placement associate with antidepressant efficacy but bilateral electrode placement may affect memory complaints post-ECT (Kolshus et al., 2017). These factors will be taken into consideration when matching the control group with the ED patient group in the current study.

2. Material and methods

2.1. Data source

Data was acquired from the Swedish National Quality Register for ECT (Q-ECT) (Ahmad et al., 2022), the National Patient Register, and the National Prescribed Drug Register. Patients treated with ECT between 2012 and 2023 were analyzed. Inclusion criteria were ED according to ICD-10 (F50.0–F50.9), unipolar depression (F32.1–F32.3, F33.1–F33.3), and evaluation with the Clinical Global Impression Improvement scale (CGI-I) within one week after the last treatment session. The control group was patients treated with ECT for unipolar depression without ED comorbidity. If a patient received several ECT treatments, the first registered series was analyzed in the study.

Prevalence of psychiatric comorbidity before ECT was extracted from the Swedish National Patient Register and included the following: eating disorder (ICD-10 code: F50), anxiety disorder (F41), personality disorder (F60–F62), obsessive-compulsive disorder (F42), autism spectrum disorder (F84.5), attention deficit hyperactivity disorder (ADHD) (F90.0) and substance use disorders (F10–F14, F16, and F18). Information about previous medication with antidepressants (ATC code: N06A), antipsychotics (N05A), and anticonvulsants (N03A) was extracted from the National Prescribed Drug Register.

2.2. Outcome measures

The primary outcome was response to treatment according to CGI-I score. The CGI-I scale assesses improvement after treatment and is clinician-rated. It ranges from scores 1 (‘very much improved’) to 7 (‘very much worse’) (Guy, 1976), where 1 was defined as remission and 1–2 as a response (Riedel et al., 2010).

The secondary outcome was a worsening of memory complaint after ECT, using the memory item of the Comprehensive Psychopathological Rating Scale (CPRS-M) which is used in the Q-ECT. The item rates subjective memory disturbance from 0 (‘memory as usual’) to 6 (‘complaints of complete inability to remember’) (Stengård et al., 2020), and is rated by the patient within one week before and after treatment. A deterioration in CPRS-M score of 2 or more was considered a subjective memory complaint.

2.3. Statistical methods

Data was managed and analyzed in the Statistical Analysis System (SAS) Enterprise Guide 9.4. The two groups were first matched on sex. Propensity score matching was then used for age, depression with psychotic features (F32.3 or F33.3), Clinical Global Impression Severity score (CGI-S) before ECT, psychiatric comorbidity with personality disorder, substance use, autism, ADHD or anxiety, number of different antidepressants before ECT, treatment with anticonvulsants and electrode placement. Two controls were matched to each case in the ED group (2:1 ratio), using matching without replacement and an optimal matching algorithm with a maximum 0.2 caliper width in propensity score logit. This caliper width has been recommended as it minimizes the mean squared error of the estimated treatment effect (Austin, 2011). The standardized mean difference (SMD) between the groups before and after matching was calculated. We set the cutoff at 0.1 or 10 %, as this has been shown to represent a negligible difference in previous research (Austin, 2009). Standardized mean difference and variance are the two most common diagnostics to assess balance following propensity score matching (Zhang et al., 2019). Frequencies of CGI-I response or remission and subjective memory complaints were calculated in the ED and control group. Logistic regression analyses were performed to compare the groups. The null hypothesis was rejected at P -values of <0.05 .

2.4. Ethical considerations

Diagnosis and treatment are sensitive personal data. Participation in Q-ECT is voluntary, and patients were informed that the data may be used for research. Personal data was pseudonymized when extracted from the register. This study is part of the research project ‘Outcome of treatment for severe affective disorders’, which has approval by the Swedish Ethical Review Authority (2023-07334-02).

3. Results

3.1. Study population

Before matching, the ED group consisted of 889 patients and the control group of 19,854. The largest SMD values were seen for sex, age, and number of antidepressant medications before ECT. After matching, the number of patients was 861 in the ED group and 1722 in the control group. Of these, 93.2% were women and 6.8% were male in both groups. The SMD value for all the included variables decreased after matching to <0.1 , except for OCD and lithium (Table 1). Note that these variables were not included in the matching procedure. The most common ED type in the population was unspecified ED, which made up 63.2%, followed by anorexia nervosa (18.3%) and bulimia nervosa (10.8%) (see Supplementary Table 1). The 28 unmatched cases were significantly more likely to be on antiepileptics ($\chi^2(1) = 5.84$, $P = 0.016$) and significantly less likely to have an ADHD diagnosis ($\chi^2(1) = 4.22$, $P = 0.040$), compared to the cases which were matched.

3.2. Primary and secondary outcomes

In the ED group, response and remission to ECT were 58.3% and 12.9%, respectively. Both outcomes were significantly lower in the ED group than in the control group (OR 0.8, 95% CI 0.7–0.9, $p < 0.01$ and OR 0.7, 95% CI 0.5–0.8, $p < 0.01$). However, before matching these differences were even more pronounced (Table 2). The reports of subjective memory complaints did not differ significantly after matching between groups (OR 1.1, 95% CI 0.8–1.3, $p = 0.64$), but were significant before matching (OR 1.7, 95% CI 1.4–2.1, $p < 0.01$) (Table 2). Comparison of response and remission for different ED types to ECT showed nonsignificant results (Table S2).

Table 1
Study population characteristics.

Characteristic	Before matching			After matching		
	ED group (n = 889)	Control group (n = 19,854)	SMD	ED group (n = 861)	Control group (n = 1722)	SMD
Sex, female, N (%)	827 (93.0)	11,778 (59.3)	0.87	802 (93.2)	1604 (93.2)	<0.01
Age, M (SD)	35.3 (13.5)	56.4 (18.6)	−1.30	35.3 (13.5)	35.8 (14.0)	−0.03
MDD with psychotic symptoms, N (%)	99 (11.1)	4660 (24.5)	−0.32	98 (11.4)	187 (10.9)	0.01
CGI-S before ECT, M (SD)	5.2 (0.9)	5.1 (0.8)	0.05	5.2 (0.9)	5.1 (0.8)	0.03
Psychiatric comorbidity, N (%)						
Anxiety disorder	465 (52.3)	8324 (41.9)	0.20	446 (51.8)	952 (55.3)	−0.07
OCD	61 (6.9)	422 (2.1)	0.18	59 (6.9)	34 (2.0)	0.18
Autism	46 (5.2)	214 (1.1)	0.23	43 (5.0)	63 (3.7)	0.08
ADHD	117 (13.2)	639 (3.2)	0.38	116 (13.5)	219 (12.7)	0.03
Personality disorder	123 (13.8)	789 (4.0)	0.36	121 (14.1)	271 (15.7)	−0.06
Substance use	109 (12.3)	2025 (10.2)	0.06	105 (12.2)	189 (11.0)	0.04
Treatment prior to ECT, N (%)						
Antidepressants	883 (99.3)	19,042 (95.9)	0.21	855 (99.3)	1696 (98.5)	0.07
Lithium	253 (28.6)	4088 (21.2)	0.15	245 (28.6)	617 (36.3)	−0.14
Benzodiazepine	738 (83.0)	14,957 (75.3)	0.16	716 (83.2)	1366 (79.3)	0.08
Antipsychotics	757 (85.2)	13,125 (66.1)	0.39	732 (85.0)	1417 (82.3)	0.06
Anticonvulsants	482 (54.2)	5105 (25.7)	0.60	464 (53.9)	952 (55.3)	−0.03
Number of medications, M (SD)	7.4 (3.0)	5.6 (3.0)	0.63	7.4 (2.9)	7.4 (3.1)	−0.02
ECT						
Number of sessions, M (SD)	7.8 (3.4)	7.6 (3.5)	0.04	7.8 (3.3)	7.7 (3.5)	0.02
Charge, M (SD)	284.5	371.8	−0.51	284.3	283.7	<0.01
Unilateral electrode placement, N (%)	801 (90.7)	17,481 (88.4)	0.06	775 (90.6)	1569 (91.8)	−0.03

Note. The threshold for the Standardized mean difference (SMD) were set at 0.1 or 10 percent, as this has been shown to equal a negligible difference in previous research by Austin (2009) and Zhang with colleagues (2019). ED = eating disorder, SMD = standardized mean difference, M = mean, SD = standard deviation, MDD = major depressive disorder, CGI-S = clinical global impression severity, OCD = obsessive-compulsive disorder, ADHD = attention deficit hyperactivity disorder, ECT = electroconvulsive therapy.

4. Discussion

This was one of the largest studies regarding the treatment of MDD for ED patients yet. It is also one of the few studies to include all ED types. Our study highlights some important findings. Firstly, a high response rate of 58.3% was observed after ECT for MDD in the ED group. This aligns with previous studies in the area, where the alleviation of depressive symptoms has been reported after ECT in ED patients (Andersson et al., 2023; Poutanen et al., 2009; Shilton et al., 2020; van der Lelie et al., 2021). This is an important finding for this patient group since pharmacological studies have shown little to no improvement from antidepressant medication in ED patients (Cassoli et al., 2020; Leblé et al., 2017; Mischoulon et al., 2011).

Table 2
Clinician-rated treatment improvement and subjective memory complaint post-ECT. Logistic regression using the control group as reference.

	CGI-I, <i>N</i> (%)		Subjective memory complaints, <i>N</i> (%)
	Response	Remission	
Before matching			
ED group	518 (58.3)	115 (12.9)	145 (26.2)
Control group	15,344 (77.3)	5647 (28.4)	2163 (17.2)
OR (95% CI)	0.4 (0.4–0.5)	0.4 (0.3–0.5)	1.7 (1.4–2.1)
<i>P</i> -value	<0.01	<0.01	<0.01
After matching			
ED group	502 (58.3)	111 (12.9)	141 (25.9)
Control group	1105 (64.2)	314 (18.2)	273 (24.8)
OR (95% CI)	0.8 (0.7–0.9)	0.7 (0.5–0.8)	1.1 (0.8–1.3)
<i>P</i> -value	<0.01	<0.01	0.64

CGI-I = clinical global impression improvement, ED = eating disorder, OR = odds ratio, CI = confidence interval.

This study also showed that subjective memory complaints were similar in ED patients and controls after matching. There was a significant difference before the groups matching, that did not remain after matching. This suggests that it is not the ED that is associated with subjective memory complaints but other patient characteristics such as age, sex, associated comorbidities, medications, or other factors. Shilton et al. also reported low frequencies of subjective memory complaints among ED patients treated with ECT (Shilton et al., 2020). Conclusively, these findings support that ECT can be an effective treatment for MDD in the ED patient group that also seems relatively well tolerated. Further studies are required to ensure efficacy and tolerability, such as randomized controlled trials versus placebo or comparisons to pharmacological treatment.

The observed response and remission rates in the matched groups were relatively low in this study. The difference between the unmatched and the matched control groups was most prominent. This can be explained by some characteristics of the ED patient group, such as younger age, lower frequency of MDD with psychotic symptoms, and higher frequency of antidepressant medication before ECT. These factors have previously been associated with poorer outcomes from ECT for MDD in several studies (Diermen et al., 2018; O'Connor et al., 2001; Petrides et al., 2001). Moreover, the number of sessions in the current study may seem somewhat low. Clinically, a treatment is ended when a patient no longer is considered to improve from ECT. Although the mean number observed still is within the range of 8–12 sessions that has previously been suggested (American Psychiatric Association, 2001), it may still partly explain the low response and remission rates reported.

After matching, a small difference in response and remission remained between the ED and control groups. Although there may be other unknown factors that alter the outcome, this finding indicates that MDD with comorbid ED is associated with poorer outcomes after ECT. This finding is similar to ECT outcome for depression with other psychiatric comorbidities, such as anxiety, substance use, and personality disorders, which have also been found associated with lower remission rates after ECT for MDD (Brus et al., 2017; Haq et al., 2015; Ferrea et al., 2024).

4.1. Limitations

Because the study design is retrospective, there are some variables we were not able to control for. One of these variables is the duration of the current depressive episode since this information is not available in the Q-ECT. Instead, the number of antidepressant medications before ECT was used as a surrogate marker. Another limitation was the use of the CGI-I scale since it assesses overall disease improvement. Therefore,

even though there was an improvement in MDD symptoms, residual ED symptoms could have affected the assessment. The overlap between symptoms seen during MDD and ED (Leblé et al., 2017) could also have complicated the evaluation of MDD. The usage of a mood scale, such as MADRS, could potentially give a better insight into patients' depressive symptoms.

In addition, the current study may suffer from selection bias, which is an inherent risk in all retrospective register studies, as the design does not allow for randomization of the exposure (eating disorder). However, the propensity score matching procedure aimed to alleviate this limitation by balancing the case group and control group on potential confounders and thereby creating a pseudo-randomization (although it cannot control for unmeasured confounders, such as a true randomization may). Lastly, future studies with a bigger sample size and usage of other measurement methods could provide more information, such as the outcomes for different ED types.

5. Conclusions

This study observed high response rates after ECT for MDD in ED patients. However, MDD with ED comorbidity was associated with lower response and remission rates compared to controls. Reports of subjective memory complaints did not differ significantly between the matched groups. ECT shows promise in treating MDD in ED patients, but additional studies are required to evaluate the effects of ECT in patients with ED complicated by MDD.

CRedit authorship contribution statement

Cecilia Gillving: Writing – original draft, Visualization, Project administration, Methodology. **Tor Arnison:** Writing – review & editing, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Axel Nordenskjöld:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

None.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jpsychires.2025.02.042>.

References

- Ahmad, I., Sandberg, M., Brus, O., Ekman, C.J., Hammar, Å., Landén, M., Lundberg, J., Nordanskog, P., von Knorring, L., Nordenskjöld, A., 2022. Validity of diagnoses, treatment dates, and rating scales in the Swedish national quality register for electroconvulsive therapy. *Nord. J. Psychiatr.* 76, 96–103. <https://doi.org/10.1080/08039488.2021.1939416>.
- American Psychiatric Association, 2013. *Diagnostic and Statistical Manual of Mental Disorders (DSM-V)*, fifth ed. American Psychiatric Publishing, Washington DC.
- American Psychiatric Association, 2001. *The Practice of ECT: Recommendations for Treatment, Training and Privileging: A Task Force on Electroconvulsive Therapy*, p. 355. Washington, DC.
- Andersson, P., Jamshidi, E., Ekman, C.-J., Tedroff, K., Björkander, J., Sjögren, M., Lundberg, J., Jokinen, J., Desai Boström, A.E., 2023. Anorexia nervosa with comorbid severe depression: a systematic scoping review of brain stimulation treatments. *J. ECT* 39, 227–234. <https://doi.org/10.1097/YCT.0000000000000922>.
- Arcelus, J., Mitchell, A.J., Wales, J., Nielsen, S., 2011. Mortality rates in patients with anorexia nervosa and other eating disorders: a meta-analysis of 36 studies. *Arch. Gen. Psychiatry* 68, 724–731. <https://doi.org/10.1001/archgenpsychiatry.2011.74>.
- Austin, P.C., 2009. Using the standardized difference to compare the prevalence of a binary variable between two groups in observational research. *Commun. Stat. Simulat. Comput.* 38, 1228–1234. <https://doi.org/10.1080/03610910902859574>.
- Blinder, B.J., Cumella, E.J., Sanathara, V.A., 2006. Psychiatric comorbidities of female inpatients with eating disorders. *Psychosom. Med.* 68, 454. <https://doi.org/10.1097/01.psy.0000221254.77675.f5>.
- Brus, O., Cao, Y., Gustafsson, E., Hultén, M., Landén, M., Lundberg, J., Nordanskog, P., Nordenskjöld, A., 2017. Self-assessed remission rates after electroconvulsive therapy of depressive disorders. *Eur. Psychiatry* 45, 154–160. <https://doi.org/10.1016/j.eurpsy.2017.06.015>.
- Calvo-Rivera, M.P., Navarrete-Páez, M.I., Bodoano, I., Gutiérrez-Rojas, L., 2022. Comorbidity between anorexia nervosa and depressive disorder: a narrative review. *Psychiatry Investig* 19, 155–163. <https://doi.org/10.30773/pi.2021.0188>.
- Cassoli, E., Sensi, C., Mannucci, E., Ricca, V., Rotella, F., 2020. Pharmacological treatment of acute-phase anorexia nervosa: evidence from randomized controlled trials. *J. Psychopharmacol. (Oxf.)* 34, 864–873. <https://doi.org/10.1177/0269881120920453>.
- Crone, C., Fochtmann, L.J., Attia, E., Boland, R., Escobar, J., Fornari, V., Golden, N., Guarda, A., Jackson-Triche, M., Manzo, L., Mascolo, M., Pierce, K., Riddle, M., Seritan, A., Uniacke, B., Zucker, N., Yager, J., Craig, T.J., Hong, S.-H., Medicus, J., 2023. The American psychiatric association practice guideline for the treatment of patients with eating disorders. *Am. J. Psychiatr.* 180, 167–171. <https://doi.org/10.1176/appi.ajp.23180001>.
- Diermen, L. van, Ameen, S. van den, Kamperman, A.M., Sabbe, B.C.G., Vermeulen, T., Schijvers, D., Birkenhäger, T.K., 2018. Prediction of electroconvulsive therapy response and remission in major depression: meta-analysis. *Br. J. Psychiatry* 212, 71–80. <https://doi.org/10.1192/bjp.2017.28>.
- Ferguson, C.P., La Via, M.C., Crossan, P.J., Kaye, W.H., 1999. Are serotonin selective reuptake inhibitors effective in underweight anorexia nervosa? *Int. J. Eat. Disord.* 25, 11–17. [https://doi.org/10.1002/\(SICI\)1098-108X\(199901\)25:1<11::AID-EAT2>3.0.CO;2-U](https://doi.org/10.1002/(SICI)1098-108X(199901)25:1<11::AID-EAT2>3.0.CO;2-U).
- Ferreira, S., Petrides, G., Ehrlich-Schäfer, Y., Angst, J., Seifritz, E., Olbrich, S., Schoretsanis, G., 2024. Outcomes of electroconvulsive therapy in patients with depressive symptoms with versus without comorbid personality disorders/traits: a systematic review and meta-analysis. *Acta Psychiatr. Scand.* 149, 18–32. <https://doi.org/10.1111/acps.13631>.
- Franko, D.L., Keel, P.K., 2006. Suicidality in eating disorders: occurrence, correlates, and clinical implications. *Clin. Psychol. Rev.* 26, 769–782. <https://doi.org/10.1016/j.cpr.2006.04.001>.
- Galmiche, M., Déchellotte, P., Lambert, G., Tavolacci, M.P., 2019. Prevalence of eating disorders over the 2000–2018 period: a systematic literature review. *Am. J. Clin. Nutr.* 109, 1402–1413. <https://doi.org/10.1093/ajcn/nqy342>.
- Gillving, C., Ekman, C.J., Hammar, Å., Landén, M., Lundberg, J., Movahed Rad, P., Nordanskog, P., von Knorring, L., Nordenskjöld, A., 2024. Seizure duration and electroconvulsive therapy in major depressive disorder. *JAMA Netw. Open* 7, e2422738. <https://doi.org/10.1001/jamanetworkopen.2024.22738>.
- Goldstein, A., Gvion, Y., 2019. Socio-demographic and psychological risk factors for suicidal behavior among individuals with anorexia and bulimia nervosa: a systematic review. *J. Affect. Disord.* 245, 1149–1167. <https://doi.org/10.1016/j.jad.2018.12.015>.
- Guy, W., 1976. *ECDEU Assessment Manual for Psychopharmacology*. U.S. Department of Health, Education, and Welfare, Public Health Service, Alcohol, Drug Abuse, and Mental Health Administration. National Institute of Mental Health, Psychopharmacology Research Branch, Division of Extramural Research Programs, California.
- Haq, A.U., Sitzmann, A.F., Goldman, M.L., Maixner, D.F., Mickey, B.J., 2015. Response of depression to electroconvulsive therapy: a meta-analysis of clinical predictors. *J. Clin. Psychiatry* 76, 1374–1384. <https://doi.org/10.4088/JCP.14r09528>.
- Kolshus, E., Jelovac, A., McLoughlin, D.M., 2017. Bitemporal v. high-dose right unilateral electroconvulsive therapy for depression: a systematic review and meta-analysis of randomized controlled trials. *Psychol. Med.* 47, 518–530. <https://doi.org/10.1017/S0033291716002737>.
- Leblé, N., Radon, L., Rabot, M., Godart, N., 2017. Depressive symptoms during anorexia nervosa: state of the art and consequences for an appropriate use of antidepressants. *L'Encephale* 43, 62–68. <https://doi.org/10.1016/j.encep.2016.02.017>.
- Milos, G., Spindler, A., Hepp, U., Snyder, U., 2004. Suicide attempts and suicidal ideation: links with psychiatric comorbidity in eating disorder subjects. *Gen. Hosp. Psychiatry* 26, 129–135. <https://doi.org/10.1016/j.genhosppsych.2003.10.005>.
- Mischoulon, D., Eddy, K.T., Keshaviah, A., Dinescu, D., Ross, S.L., Kass, A.E., Franko, D.L., Herzog, D.B., 2011. Depression and eating disorders: treatment and course. *J. Affect. Disord.* 130, 470–477. <https://doi.org/10.1016/j.jad.2010.10.043>.
- O'Connor, M.K., Knapp, R., Husain, M., Rummans, T.A., Petrides, G., Smith, G., Mueller, M., Snyder, K., Bernstein, H., Rush, A.J., Fink, M., Kellner, C., 2001. The influence of age on the response of major depression to electroconvulsive therapy: a C.O.R.E. Report. *Am. J. Geriatr. Psychiatry Off. J. Am. Assoc. Geriatr. Psychiatry* 9, 382–390.
- Pagnin, D., de Queiroz, V., Pini, S., Cassano, G.B., 2004. Efficacy of ECT in depression: a meta-analytic review. *J. ECT* 20, 13–20. <https://doi.org/10.1097/00124509-200403000-00004>.
- Petrides, G., Fink, M., Husain, M.M., Knapp, R.G., Rush, A.J., Mueller, M., Rummans, T.A., O'Connor, K.M., Rasmussen, K.G., Bernstein, H.J., Biggs, S.H., Bailine, S.H., Kellner, C.H., 2001. ECT remission rates in psychotic versus nonpsychotic depressed patients: a report from CORE. *J. ECT* 17, 244–253. <https://doi.org/10.1097/00124509-200112000-00003>.

- Poutanen, O., Huuhka, K., Perko, K., 2009. Severe anorexia nervosa, co-occurring major depressive disorder and electroconvulsive therapy as maintenance treatment: a case report. *Cases J* 2, 9362. <https://doi.org/10.1186/1757-1626-2-9362>.
- Riedel, M., Möller, H.-J., Obermeier, M., Schennach-Wolff, R., Bauer, M., Adli, M., Kronmüller, K., Nickel, T., Brieger, P., Laux, G., Bender, W., Heuser, I., Zeiler, J., Gaebel, W., Seemüller, F., 2010. Response and remission criteria in major depression – a validation of current practice. *J. Psychiatr. Res.* 44, 1063–1068. <https://doi.org/10.1016/j.jpsychires.2010.03.006>.
- Shilton, T., Enoch-Levy, A., Giron, Y., Yaroslavsky, A., Amiaz, R., Gothelf, D., Weizman, A., Stein, D., 2020. A retrospective case series of electroconvulsive therapy in the management of comorbid depression and anorexia nervosa. *Int. J. Eat. Disord.* 53, 210–218. <https://doi.org/10.1002/eat.23181>.
- Silén, Y., Keski-Rahkonen, A., 2022. Worldwide prevalence of DSM-5 eating disorders among young people. *Curr. Opin. Psychiatr.* 35, 362–371. <https://doi.org/10.1097/YCO.0000000000000818>.
- Stengård, J., Ekman, C.J., Båve, U., Lundahl, A., Abawi, M., Adler, M., Landén, M., Lundberg, J., 2020. Physician estimated vs. self-reported subjective memory in depressed patients treated with electroconvulsive therapy. *Nord. J. Psychiatr.* 74, 359–365. <https://doi.org/10.1080/08039488.2020.1719194>.
- UK ECT Review Group, 2003. Efficacy and safety of electroconvulsive therapy in depressive disorders: a systematic review and meta-analysis. *Lancet Lond. Engl.* 361, 799–808. [https://doi.org/10.1016/S0140-6736\(03\)12705-5](https://doi.org/10.1016/S0140-6736(03)12705-5).
- van der Lelie, K., Sienaert, P., Vrieze, E., 2021. Electroconvulsive therapy in the treatment of depression in anorexia nervosa. *Tijdschr. Psychiatr.* 63, 673–678.
- Zhang, Z., Kim, H.J., Lonjon, G., Zhu, Y., 2019. Balance diagnostics after propensity score matching. *Ann. Transl. Med.* 7. <https://doi.org/10.21037/atm.2018.12.10>, 16–16.